



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

ISE-I (Semester-I) 2025-26

Second Year B. Tech.

Civil Engineering

Subject: Building Construction

Subject Code: PCC-CE204

Date: 19-Aug-2025

Time: 45 Minutes (11:00 to 11:45)

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	Explain the different types of shallow and deep foundations with neat sketches? State the selection criteria of foundations.	10	CO2	BT2
Q. 2)	State the various types of doors and windows with neat sketches.	10	CO1	BT1



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded.
Department of Civil Engineering ISE-I Examination Academic Year: 2025-26 (Semester I)

Class: S.Y. B. Tech. Subject: Fluid Mechanics & Fluid Mechanics-I Subject Code: PCC-CE202/CE203 & CW202
Date: 21/08/2025 Time: 11:00 a.m. to 11:45 p.m. Max. Marks: 20

Note: i. Assume suitable data whenever necessary and state it clearly. ii. Mobile phones and programmable calculators are strictly prohibited. iii. Exchange/Sharing of stationery, calculator etc. not allowed. iv. Figures to right indicate full marks. v. The symbols carry usual meaning.

Q. No.	Question	Marks	CO	BT
Q.1 (a)	What is meant by specific gravity of a fluid? The surface tension of water at 20°C is 0.072 N/m . The pressure inside a water droplet is to be 0.03 N/cm^2 greater than outside pressure. Calculate the diameter of a water droplet.	5	CO1	BT1
Q.1 (b)	Define the following: (1) Dynamic viscosity of fluid (2) Vapour pressure. A 6 m deep tank contains 4 m of water and 2 m of oil of relative density 0.88. Determine the pressure at the bottom of the tank.	5	CO1	BT2
Q.2 (a)	Define the following: (1) Buoyant force (2) Metacentre. The pressure intensity at a point in the fluid is 5 N/cm^2 . Find the corresponding height of columns of water and oil of specific gravity 0.88.	5	CO1	BT1
Q.2 (b)	Define the following: (1) Velocity potential (2) Stream function. For the stream function $\psi = 3x^2y - y^3$ calculate the velocity at a point (2, 3).	5	CO2	BT2

Bloom's Taxonomy: (BT):BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create
Course Outcomes: On successful completion of this course, students should be able to:
CO 1: Understand fluid properties and fluid statics principles CO 2: Understand fluid kinematics principles

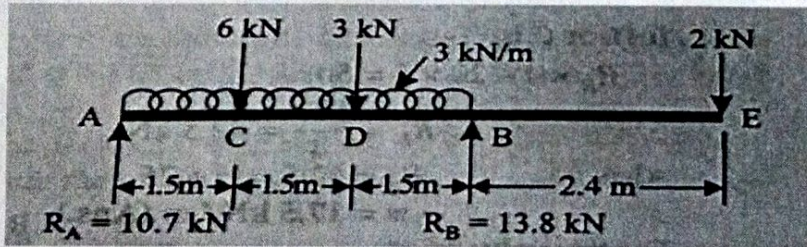


Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
In-Semester Examination 1
Academic Year: 2025-26 (Semester – I)

Class: SY	Subject: Strength of Materials	Subject Code: PCC-CE201
Date: 22.08.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Mobile phones and programmable calculators are strictly prohibited.
- ii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.
- v. All questions are compulsory

Q.N.			Marks	CO	BT
Q.1	a	Explain the following terms 1) Stress-strain curve for mild steel 2) Concept of temperature stresses for bimetallic strips	6	CO1	BT1
Q.2	a	The following data relate to a bar subjected to a tensile test: Diameter of the bar, $d=32$ mm Tensile load, $P=50$ kN Gauge length, $l=300$ mm Extension of the bar, $\delta l=0.212$ mm Change in diameter, $\delta d=0.00466$ mm Calculate (i) Poisson's ratio (ii) The values of the three elastic moduli (give proper units)	7	CO2	BT4
Q.3	a	Draw Shear force diagram (SFD) and Bending moment diagram (BMD) for the following beam. (give proper units) 	7	CO5	BT5

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Recall stress and strain fundamentals, Hooke's law, principal stresses, strain energy, and column buckling.
CO 2	Understand how load intensity, stress, strain, bending moment, and shear force influence structural member behaviour under different loads.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Civil Engineering

Summer Term Examination

Academic Year: 2024-25

Class: SY B.Tech.
(Common to All)

Course: Project Management

Course Code: HSSM201

Date: 19.06.2025

Time: 14:30 to 17:30

Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT																						
Q.1	a	Define project and explain the basic characteristics of a project.	7	CO1	BT1																						
	b	List various types of project appraisal. Explain any two.	8	CO2	BT1																						
Q.2	a	Explain the Gantt chart and its role in project scheduling.	6	CO2	BT2																						
	b	Construct the network diagram for the following project: <table border="1"><thead><tr><th>Activity</th><th>Immediate Predecessor Activity</th></tr></thead><tbody><tr><td>A</td><td>-</td></tr><tr><td>B</td><td>-</td></tr><tr><td>C</td><td>A</td></tr><tr><td>D</td><td>B</td></tr><tr><td>E</td><td>A</td></tr><tr><td>F</td><td>C, D</td></tr><tr><td>G</td><td>E</td></tr><tr><td>H</td><td>E</td></tr><tr><td>I</td><td>F, G</td></tr><tr><td>J</td><td>H, I</td></tr></tbody></table>	Activity	Immediate Predecessor Activity	A	-	B	-	C	A	D	B	E	A	F	C, D	G	E	H	E	I	F, G	J	H, I	6	CO2	BT3
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Q.3	a	Write short notes on: (a) Project crashing (b) Resource leveling	6	CO3	BT2																						
	b	Find out the completion time and the critical activities for the following project: (fig.01)	6	CO4	BT3																						

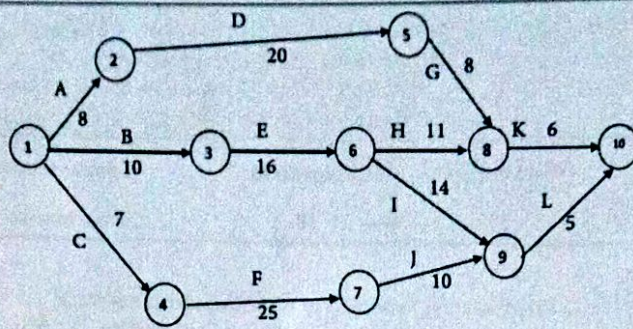


Fig.01

Fig.01

Q.4	a	What are the responsibilities of a project manager?	8	CO3	BT2																																							
	b	Describe the role of project planning in software development.	6	CO2	BT2																																							
Q.5	a	Draw the network diagram for the following project. Determine the time, variance and standard deviation of the project:	11	CO4	BT3																																							
		Activity				Predecessor Activity	Optimistic estimate of time	Most likely estimate of time	Pessimistic estimate of time	A	-	12	14	22	B	-	16	17	24	C	A	14	15	16	D	A	13	18	23	E	B	16	18	20	F	D,E	13	14	21	G	C,F	6	8	10
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G	C,F	6	8	10																																								
b	Explain the concept of crashing of a project.	06	CO3	BT2																																								
Q.6	a	Consider a project with the following details:	15	CO4	BT4																																							

		<table><tr><th>Name of Activity</th><th>Predecessor Activity</th><th>Duration (Weeks)</th></tr><tr><td>A</td><td>-</td><td>8</td></tr><tr><td>B</td><td>A</td><td>13</td></tr><tr><td>C</td><td>A</td><td>9</td></tr><tr><td>D</td><td>A</td><td>12</td></tr><tr><td>E</td><td>B</td><td>14</td></tr><tr><td>F</td><td>B</td><td>8</td></tr><tr><td>G</td><td>D</td><td>7</td></tr><tr><td>H</td><td>C, E, G</td><td>12</td></tr><tr><td>I</td><td>C, E, G</td><td>9</td></tr><tr><td>J</td><td>E, H</td><td>10</td></tr><tr><td>K</td><td>I, J</td><td>7</td></tr></table> Determine the earliest and latest times, the total float for each activity, the critical activities, the slacks of the events and the project completion time.	Name of Activity	Predecessor Activity	Duration (Weeks)	A	-	8	B	A	13	C	A	9	D	A	12	E	B	14	F	B	8	G	D	7	H	C, E, G	12	I	C, E, G	9	J	E, H	10	K	I, J	7			
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Q.7	a	Define Payback Period and Benefit-Cost Ratio. Explain with examples.	8	CO1	BT1																																				
	b	Illustrate the use of Microsoft Project software for time-cost optimization.	7	CO2	BT3																																				

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Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Appraise the selection and initiation of individual projects and its portfolios in an enterprise.
CO 2	Analyze the project planning activities that will predict project costs, time schedule, and quality.
CO 3	Develop processes for successful resource allocation, communication, and risk management.
CO 4	Evaluate effective project execution and control techniques that results in successful project completion.